fsstat Forensic Report

1. General File System Information

File System Type: NTFS

Volume Serial Number: 4C7E1AB47E1A96B4

OEM Name: NTFS

Volume Name: New Volume0

Version: Windows XP

Indicates the type of file system (e.g., NTFS). This is essential for understanding the structure and forensic artifacts specific to the file system. A unique identifier for the volume. It helps in matching volumes during multi-disk forensic investigations. Represents the OEM name of the file system (typically the same as the file system type for NTFS). Useful for verifying the file system’s origin. The label assigned to the volume. It can be used to quickly identify the partition in investigations. The version of the file system. For NTFS, this can help in determining compatibility and potential forensic analysis nuances.

2. Metadata Information

First Cluster of MFT: 8533

First Cluster of MFT Mirror: 2

Size of MFT Entries: 1024 bytes

Size of Index Records: 4096 bytes

Range: 0 - 256

Root Directory: 5

Indicates where the Master File Table (MFT) begins. The MFT is central to NTFS and crucial for file record analysis. Specifies the location of the MFT mirror, which is used for redundancy. Its location helps verify the integrity of the MFT. Shows the size of each MFT entry. Understanding this helps in interpreting file records stored in the MFT. Indicates the size of the index records used in the file system. This is important for analyzing the structure of directories and indexes. Provides a range related to metadata information. This may denote limits or counts relevant to metadata entries. Specifies the root directory entry. This is key for navigating the file hierarchy in NTFS.

3. Content Information

Sector Size: 512 bytes

Cluster Size: 4096 bytes

Total Cluster Range: 0 - 25598

Total Sector Range: 0 - 204798

The size of a sector on the disk. It is fundamental for understanding disk geometry and storage allocation. The size of a cluster, which is a group of sectors. It influences how data is stored and how much slack space may exist. Specifies the range of clusters in the partition. This helps in determining the partition's capacity and structure. Specifies the range of sectors in the partition. This is important for mapping the physical layout of the disk.

4. NTFS Attribute Definitions ($AttrDef)

Attribute	Identifier	Size Range	Flags	Description
$STANDARD_INFORMATION	16	48-72	Resident	Contains standard metadata for files, including timestamps and file permissions. The flags indicate that this attribute is stored within the MFT entry (resident).
$ATTRIBUTE_LIST	32	No Limit	Non-resident	Holds a list of attributes when there are too many to fit in a single MFT entry. Non-resident indicates that the attribute's data is stored outside the MFT entry.
$FILE_NAME	48	68-578	Resident, Index	Stores the file name and related information. The 'Index' flag indicates it may be used in directory indexing.
$OBJECT_ID	64	0-256	Resident	Used for object identification, providing a unique identifier for files. Typically stored resident in the MFT entry.
$SECURITY_DESCRIPTOR	80	No Limit	Non-resident	Contains security information such as access control lists (ACLs). Its non-resident nature means the descriptor is stored outside the MFT entry.
$VOLUME_NAME	96	2-256	Resident	Stores the volume label of the partition. It helps in identifying the volume.
$VOLUME_INFORMATION	112	12-12	Resident	Provides additional volume-specific information, stored resident in the MFT.
$DATA	128	No Limit		Holds the actual file data. Its size being 'No Limit' indicates that the data is stored separately from the MFT entry.
$INDEX_ROOT	144	No Limit	Resident	Defines the root of a directory index, which is used to organize directory contents.
$INDEX_ALLOCATION	160	No Limit	Non-resident	Contains additional index entries that do not fit within the $INDEX_ROOT attribute. Being non-resident, it is stored outside the MFT entry.
$BITMAP	176	No Limit	Non-resident	Used to track which clusters are in use. A non-resident attribute stored separately from the MFT.
$REPARSE_POINT	192	0-16384	Non-resident	Contains reparse point data, which is used for directory junctions and symbolic links.
$EA_INFORMATION	208	8-8	Resident	Holds information about extended attributes (EA), stored resident in the MFT.
$EA	224	0-65536		Contains the actual extended attribute data. Its size can vary widely depending on the EA content.
$LOGGED_UTILITY_	256	0-65536	Non-resident	Used for logging purposes by the file system, stored non-resident.

Attribute	Identifier	Size Range	Flags	Description
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5. Overall Significance

The provided file system information, metadata details, and content layout offer critical insights into the disk partition's structure and health. The NTFS attribute definitions further enhance the forensic analysis by clarifying how file-related data is stored and managed. Understanding these details helps investigators identify the partition, interpret file system artifacts, and prioritize areas for deeper analysis in forensic investigations.